

(base) sorenbasnet@Sorens-MacBook-Pro-2 datafiles % ./guide < input.txt
GUIDE Classification and Regression Trees and Forests
Version 46.1 (Build date: January 23, 2026)
Compiled with NAG Fortran 7.2 on macOS Sequoia 15.6.1 for Apple ARM processors
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National Science Foundation, National Institutes of Health,
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Choose one of the following options:

0. Read the warranty disclaimer

1. Create a GUIDE input file

Input your choice: Batch run with input file

Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data conversion: 1

Job start date and time: 01/28/26 at 10:43

Input 1 for single tree, 2 for ensemble of trees: 1

Input 1 for classification, 2 for regression, 3 for propensity score tree

Input your choice: 1

Input 1 for simple, 2 for nearest-neighbor, 3 for kernel, 4 for lda models, 5 for qda models: 1

Input 0 for linear, interaction and univariate splits (in this order),

1 for univariate, linear and interaction splits (in this order),

2 to skip linear splits,

3 to skip linear and interaction splits: 1

Input 0 to specify tree with fixed no. of nodes, 1 to prune by CV,

2 by test sample, 3 for no pruning: 1

Bonferroni alpha = 0.100

Input name of DSC file (max 100 characters);

enclose with matching quotes if it has spaces: body.dsc

Reading DSC file ...

Training sample file: bodydata.txt

Missing value code: NA

Records in data file start on line 2

24 N variables changed to S

D variable is Gender

Reading data file ...

Number of records in data file: 507

Length of longest entry in data file: 5

Checking for missing values ...

Finished checking

Recoding D values to integers

Finished recoding

Number of classes: 2

Re-checking data ...

Assigning codes to missing values, if any ...

Data checks complete

Rereading data ...

Class #Cases Proportion

F 260 0.51282051

M 247 0.48717949

Numbers of N, F, S, and P below exclude constant variables

Total #cases w/ #missing

#cases miss. D ord. vals #X-var #N-var #F-var #S-var

507 0 0 0 0 0 24

#P-var #M-var #B-var #C-var #I-var

0 0 0 0 0

Number of cases used for training: 507

Finished reading data file

Input number of cross-validations: 10

Selected tree is based on mean of CV estimates

Input number of SEs for pruning: 0.250

Choose 1 for estimated priors, 2 for equal priors, 3 to input priors from a file

Input 1, 2, or 3: 1

Choose 1 for unit misclassification costs, 2 to input costs from a file

Input 1 or 2: 1

Univariate split highest priority

Interaction and linear splits 2nd and 3rd priorities

Choose a split point selection method for numerical variables:

Choose 1 to use faster method based on sample quantiles

Choose 2 to use exhaustive search

Input 1 or 2: 2

Maximum number of split levels: 15

Input 1 for default minimum node size, 2 to specify minimum value: 1

Minimum node size: 2

Choose 0=no LaTeX, 1=tree without node numbers, 2=tree with node numbers: 2

Input file name to store LaTeX code: class.tex

Input 0=highest posterior, 1=sample sizes, 2=sample proportions, 3=posterior probs, 4=nothing:

3

You can store the variables and/or values used to split and fit in a file

Choose 1 to skip this step, 2 to store split and fit variables,

3 to store split variables and their values

Input your choice: 1

Input 2 to save fitted values and node IDs; 1 otherwise: 2

File name is classfit.txt

Input 1 to skip R function, 2 to output R for training data, 3 to output R for new data file: 2

File name for R code: class.r

Constructing main tree ...

Top-10 ranked variables and their 1-df chi-squared values at root node

1	0.3097E+03	Forearmgirth
2	0.2976E+03	Shouldergirth
3	0.2947E+03	Biacromialdiam
4	0.2937E+03	Wristmingirth
5	0.2849E+03	Elbowdiam
6	0.2639E+03	Bicepgirth
7	0.2632E+03	Chestgirth
8	0.2551E+03	Wristdiam
9	0.2444E+03	Chestdiam
10	0.2286E+03	Waistgirth

Best and 2nd best split variables at root node are Forearmgirth and Shouldergirth

Number of subtrees: 7

Performing 10-fold cross-validation:

Completed cross-validation fold 1

Completed cross-validation fold 2

Completed cross-validation fold 3

Completed cross-validation fold 4

Completed cross-validation fold 5

Completed cross-validation fold 6

Completed cross-validation fold 7

Completed cross-validation fold 8

Completed cross-validation fold 9

Completed cross-validation fold 10

Pruning main tree. Please wait.

Results of subtree sequence

Trees based on mean with naive SE are marked with * and **

Tree based on mean with bootstrap SE is marked with --

Trees based on median with finite bootstrap SE are marked with + and ++

Subtree	#Terminal nodes
1	12
2**	10
3	8
4	6
5	4
6	2
7	1

0-SE tree based on mean is marked with * and has 10 terminal nodes

* tree, ** tree, + tree, and ++ tree all the same

Number of terminal nodes of final tree: 10

Writing predicted values...

...completed

Results are stored in output.txt

Observed and fitted values are stored in classfit.txt

LaTeX code for tree is in class.tex

Elapsed time in seconds: 0.4395

Warning: Floating overflow occurred